

Method of Moments and Minimum Mean Square Error (MSE) Estimators.I. Method of Moments Estimator (Based on substitution principle of Karl Pearson, 1890)

$X_1, \dots, X_n$ is a random sample from some distribution with pdf/pdf to, $\theta \in \Theta$.

We're interested in estimating some known function of a finite no. of unknown moments, i.e., let the j th population raw moment (~~the j th~~)

$\mu_j' = E(X_1^j)$ exists for $1 \leq j \leq k$ and

$\theta = h(\mu_1', \dots, \mu_k')$ for a known function $h(\cdot)$, then the

method of moments estimates θ by the statistic,

$$\hat{\theta} = T(X) = h\left(\frac{1}{n} \sum_{i=1}^n X_i, \frac{1}{n} \sum_{i=1}^n X_i^2, \dots, \frac{1}{n} \sum_{i=1}^n X_i^k\right) = h(m_1', \dots, m_k'), \text{ where } m_j' \text{ is the corresponding } j\text{th sample (raw) moment.}$$

For several such parameters of interest, let

$\theta_i = h_i(\mu_1', \dots, \mu_k')$, h_i 's are all known functions,

Method of moments estimator (MOME) of θ_i is

$$\hat{\theta}_i = h_i(m_1', \dots, m_k') = T_i(X)$$

Incidentally, $m_j' = \frac{1}{n} \sum_{i=1}^n X_i^j$ is the MOME of μ_j' , $j=1(1)k$.

Ex 1 $X_1, \dots, X_n$ are iid r.v.'s with mean μ & variance σ^2 , i.e.,

$$\left. \begin{aligned} E(X_1) = \mu_1' = \mu &\Rightarrow \mu = h_1(\mu_1', \mu_2') = \mu_1' \\ \text{Var}(X_1) = \mu_2' - \mu_1'^2 = \sigma^2 &\Rightarrow \sigma^2 = h_2(\mu_1', \mu_2') = \mu_2' - \mu_1'^2 \end{aligned} \right\}$$

Hence, MOME's of μ & σ^2 are

$$\hat{\mu} = h_1(m_1', m_2') = m_1' = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

$$\hat{\sigma}^2 = h_2(m_1', m_2') = m_2' - m_1'^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{1}{n} \sum_{i=1}^n X_i\right)^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

Ex 2 $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$.

Then as in Ex 1 $\hat{\mu} = \bar{X}$, $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$,

Also, $\sigma = \sqrt{h_2(\mu_1', \mu_2')} = h_3(\mu_1', \mu_2') = \sqrt{h_2(\mu_1', \mu_2')}$ for a known h_3 ,

$\therefore$ MOME of σ is $\hat{\sigma} = h_3(m_1', m_2') = \sqrt{h_2(m_1', m_2')} = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$, much

different from an unbiased estimator of σ .

Ex 3

$X_1, \dots, X_n$ iid from $P(\theta)$, $\theta > 0$.

Here $E(X_1) = V(X_1) = \theta$, so that both

$\bar{X}$ and $\frac{1}{n} \sum (X_i - \bar{X})^2$ are MOMEs of θ

But usually we take the estimator which uses lower order sample moments. Hence, here $\bar{X}$ is the chosen one.

Ex 4.

$X_1, \dots, X_n$ iid from $U(-\theta, \theta)$, $\theta > 0$

Note, here $\mu_1' = E(X_1) = \frac{1}{2\theta} \int_{-\theta}^{\theta} x dx = 0$

So, we need to move to the next higher ^{non-zero} moment.

$$\mu_2' = E(X_1^2) = \frac{1}{2\theta} \int_{-\theta}^{\theta} x^2 dx = \frac{\theta^2}{3} \Rightarrow \theta^2 = 3\mu_2' \text{ and } \theta = \sqrt{3\mu_2'} \text{ (-ve sign ignored)}$$

So the MOME of θ is $\hat{\theta} = \sqrt{3 \frac{1}{n} \sum X_i^2}$

Ex 5.

$X_1, \dots, X_n$ iid from $U(\theta_1, \theta_2)$, $\theta_1 < \theta_2$, $\theta_i \in \mathbb{R}$, $i=1,2$

Here, if $\theta_1 = h_1(\mu_1', \mu_2')$ and $\theta_2 = h_2(\mu_1', \mu_2')$, we need to know h_1 & h_2 completely.

$$\text{Now, } \mu_1' = E(X_1) = \frac{\theta_1 + \theta_2}{2} \rightarrow (i)$$

$$\text{and } \mu_2' - \mu_1'^2 = \text{Var}(X_1) = \frac{(\theta_2 - \theta_1)^2}{12} \rightarrow (ii)$$

$$\text{from (i), } \theta_1 = 2\mu_1' - \theta_2, \text{ putting in (ii), } \mu_2' - \mu_1'^2 = \frac{(\theta_2 - \mu_1')^2}{3}$$

$$\Rightarrow \theta_2 = \mu_1' + \sqrt{3(\mu_2' - \mu_1'^2)} = h_2(\mu_1', \mu_2')$$

$$\therefore \theta_1 = 2\mu_1' - \theta_2 = \mu_1' - \sqrt{3(\mu_2' - \mu_1'^2)} = h_1(\mu_1', \mu_2')$$

$\therefore$ MOMEs of θ_1 & θ_2 are,

$$\hat{\theta}_1 = \bar{X} - \sqrt{3 \frac{\sum (X_i - \bar{X})^2}{n}}, \quad \hat{\theta}_2 = \bar{X} + \sqrt{3 \frac{\sum (X_i - \bar{X})^2}{n}}$$

The above ~~ex~~ example shows that though the concept is simple, MOME is not popular as working it out may not be simple — complete identification of several such ~~closed form~~ functions h_i 's may not be even possible!

II: Minimum MSE Estimator

3/1

Defn Mean Square Error (MSE) of any estimator $\delta(x)$ of $g(\theta)$ is

$$MSE_{\theta}(\delta(x)) = E_{\theta}(\delta(x) - g(\theta))^2$$

$$\text{Note } MSE_{\theta}(\delta(x)) = \text{Var}_{\theta}(\delta(x)) + \text{Bias}_{\theta}^2(\delta(x)) \quad (\text{why?})$$

$$\text{and, } = \text{Var}_{\theta}(\delta(x)) \text{ if } \text{Bias}_{\theta}(\delta(x)) = 0, \\ \text{or, } E_{\theta}(\delta(x)) = g(\theta) \quad \forall \theta$$

Defn Let Δ be a class of estimators of $g(\theta)$.

A $\delta^* \in \Delta$ is the min MSE estimator of $g(\theta)$ if

$$\text{MSE}_{\theta}(\delta^*(x)) \leq \text{MSE}_{\theta}(\delta(x)) \quad \forall \theta \in \Theta, \quad \forall \delta \in \Delta \\ \text{and, } < \quad \text{for some } \theta$$

Ex $x_1, \dots, x_n$ are iid $N(\mu, \sigma^2)$ observations.

Let $\Delta = \{ \delta_c(x) : \delta_c(x) = c \sum (x_i - \bar{x})^2 \}$ be a class of estimators of $g(\sigma^2) = \sigma^2$.

$$\text{Then } \underline{MSE_{\theta}(\delta_c(x))} = E_{\theta}[\delta_c(x) - \sigma^2]^2 \\ = E_{\theta}[c S_x^2 - \sigma^2]^2, \quad c S_x^2 = c \sum (x_i - \bar{x})^2$$

$$\therefore \frac{\sum (x_i - \bar{x})^2}{\sigma^2} = \frac{S_x^2}{\sigma^2} \sim \chi_{n-1}^2, \quad E(S_x^2) = (n-1)\sigma^2, \quad \text{Var}(S_x^2) = 2\sigma^4(n-1) \\ \Rightarrow E(S_x^2)^2 = E(S_x^4) = \sigma^4(n-1)(n+1)$$

$$\text{Hence, } MSE_{\theta}(\delta_c(x)) = \sigma^4 [c^2(n-1) - 2c(n-1) + 1]$$

Minimizing w.r.t c , we find the min MSE estimator of

$$\frac{\partial MSE_{\theta}(\delta_c(x))}{\partial c} \Big|_{c^*} = 0 \Rightarrow c^* = \frac{1}{n+1} \text{ and } \delta_{c^*} = \frac{1}{n+1} \sum (x_i - \bar{x})^2 \text{ is the } \\ \underline{\text{min}} \text{ MSE estimator.}$$

⚡ Different from an unbiased estimator, the UMVUE, or even the Maximum Likelihood Estimator, (next)